

MATH FOR AI • EPISODE 0.2

Reading Math Out Loud

Greek letters, subscripts, and the big Σ

Dr. Joe

Can you read this out loud?

$$\text{Error} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

By the end of this episode, you will. And you'll know what every piece means.

Math is a language. Symbols are shorthand.

THE LONG WAY (ENGLISH)

For each student, find how far off the guess was, multiply it by itself, add those up, then divide by how many students there are.

THE SHORT WAY (MATH)

$$\text{Error} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Same instructions. Math just writes them faster. Today's rule: **say the name, then say what it does.**

Meet the Greek letters

α

alpha
"AL-fuh"

β

beta
"BAY-tuh"

δ

delta
"DEL-tuh"

ϵ

epsilon
"EP-sih-lon"

η

eta
"AY-tuh"

θ

theta
"THAY-tuh"

λ

lambda
"LAM-duh"

μ

mu
"MYOO"

σ

sigma
"SIG-muh"

Σ

Sigma
"capital sigma"

What they usually mean in AI

Letter	Name	Its usual job in AI
η	eta	step size (the learning rate) · you met it in Ep 0.1
θ	theta	all the numbers a model can adjust (like w and b)
μ	mu	an average (the mean)
σ	sigma	spread: how far things wander from average
Σ	Sigma	add them all up
ϵ	epsilon	a tiny number (or random noise)
λ	lambda	a setting you choose (how strict to be)
Δ	Delta (capital δ)	a change in something

These are habits, not laws. Always check how a paper defines its symbols.

Hats, bars, and stars

 $\hat{y}$

“y-hat”

a prediction (a guess)

Ep 0.1: our guess for student 1 was 45

 $\bar{x}$

“x-bar”

an average

A bar on any letter means its average

 w^*

“w-star”

the best value

Ep 0.1: the best slope w was
about 8.18

A hat is a guess. A bar is an average. A star is the best.

Subscripts are name tags

	x_1	x_2	x_3	x_4	x_5
hours	1	2	3	4	5
	y_1	y_2	y_3	y_4	y_5
scores	49	56	65	71	82

“y sub 3” = 65
the 3rd student’s score

“x sub 3” = 3
the 3rd student’s hours

Coincidence alert: student 3 happened to study 3 hours. The subscript is the seat number, not the value.

i = which one (it moves)

n = how many ($n = 5$)

Low is a label. High is a power.

SUBSCRIPT (LOW) = A LABEL

$$y_2 = 56$$

“y sub 2”: student 2 scored 56

A NUMBER UP HIGH = A POWER

$$3^2 = 3 \cdot 3 = 9$$

“3 squared”: multiply 3 by itself
(A star, like w^* , is just a label.)

BOTH AT ONCE

$$y_1^2 = 49^2 = 2401$$

“y sub 1, squared”: student 1’s score, times itself

Σ means: add them all up

ADD UP → $\sum_{i=1}^n x_i$

n ← STOP after $i = n$
(n is included!)

x_i ← WHAT to add
each time

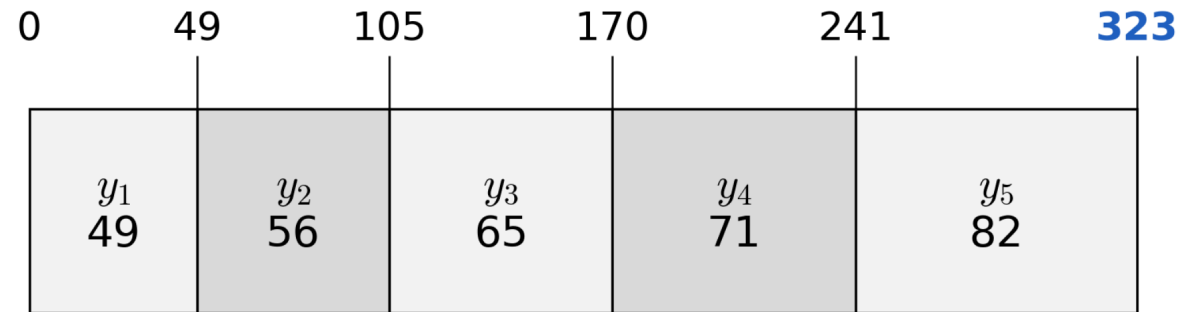
$i = 1$ ← START at $i = 1$,
then count up by 1

$$\sum_{i=1}^n x_i = x_1 + x_2 + x_3 + \cdots + x_n$$

Let's add the five scores by hand

$$\sum_{i=1}^5 y_i = 49 + 56 + 65 + 71 + 82 = 323$$

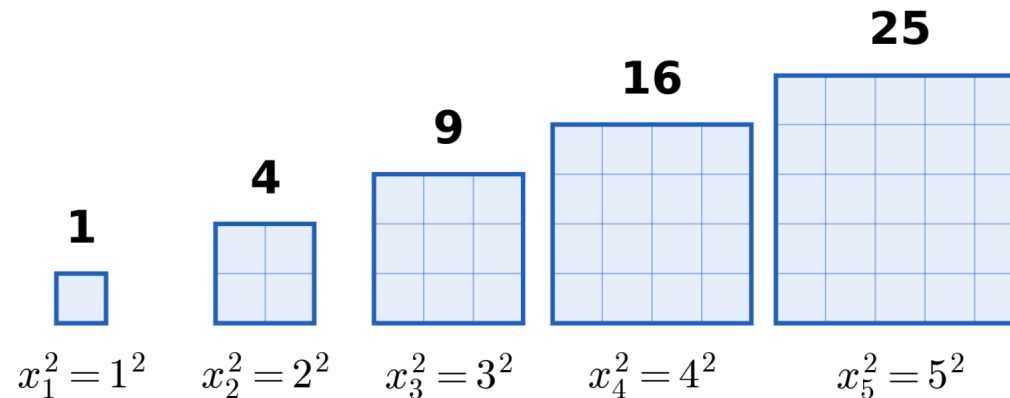
Running total: add one student at a time. The total is 323.



Change what you add: the squares

$$\sum_{i=1}^5 x_i^2 = 1 + 4 + 9 + 16 + 25 = 55$$

$1 + 4 + 9 + 16 + 25 = 55$ tiles



Careful: square first, then add = 55. Add first, then square = $15^2 = 225$. Order matters!

Σ is just a for loop

$$\sum_{i=1}^n y_i =$$

math

```
scores = [49, 56, 65, 71, 82]
n = len(scores)           # n = 5
total = 0                 # total starts as 0
for i in range(n):        # i = 0, 1, 2, 3, 4
    total = total + scores[i] # add on this score
```

Python

Watch out: Python counts from 0, so math's **y sub 1** lives at Python's **scores[0]**.

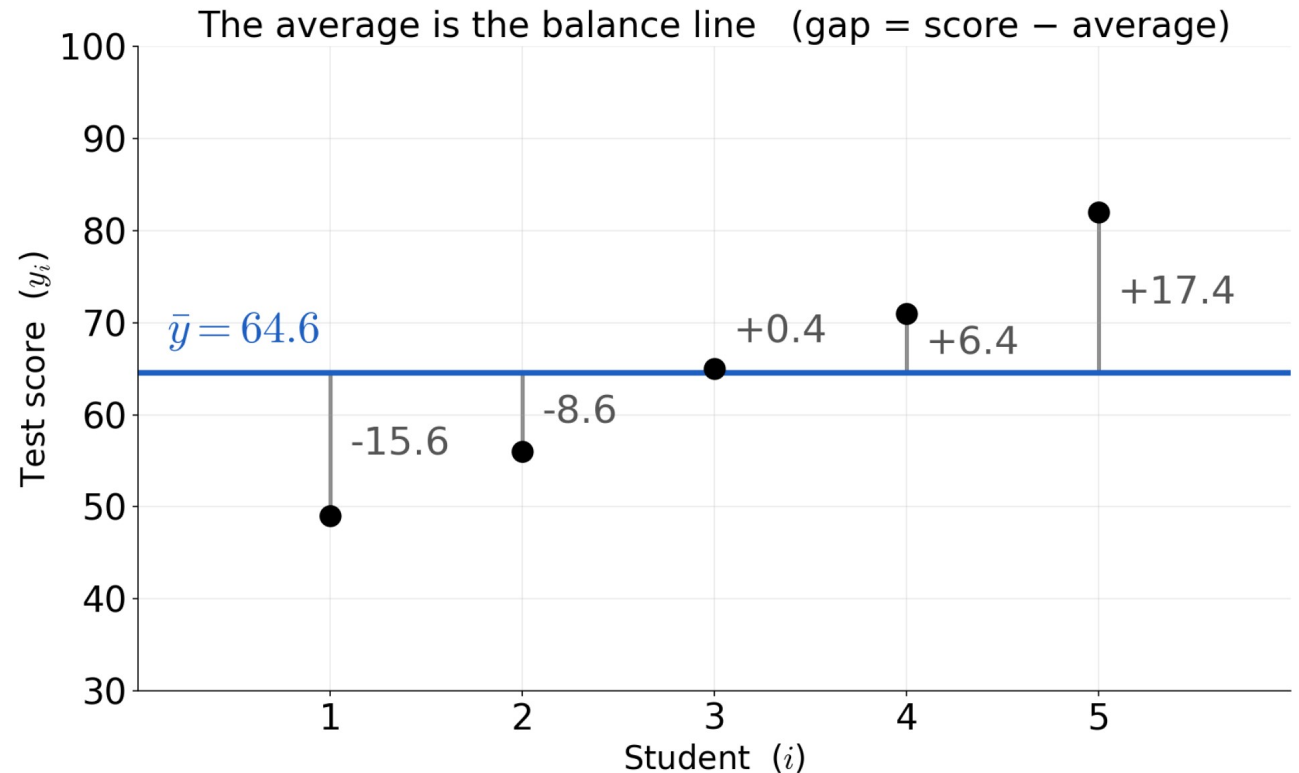
And in Python, = means “store this,” not “these are equal.”

The average is a sum, divided by n

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$\bar{y} = \frac{323}{5} = 64.6$$

Say: “y-bar equals one over n, times the sum from i equals 1 to n of y sub i.” (Times one-fifth = divide by 5.)



Now read the whole thing

$$\text{Error} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

SAY

Error equals

one over n

the sum from
i = 1 to n of

y sub i minus
y-hat sub i, squared

MEANS

how wrong
we are

divide by
how many

add up, for
every student

each student's
miss, squared

Read it left to right. Do it inside-out: **miss** → **square** → **add up** → **divide by n**.

One formula, four ways

SYMBOLS

$$\text{Error} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

OUT LOUD

“Error equals one over n times the sum from i equals 1 to n of y sub i minus y-hat sub i, squared.”

ENGLISH

Find each student’s miss, square it, add them all up, then divide by how many students.

PYTHON

```
total = 0
for i in range(n):
    y_hat = 5 * hours[i] + 40          # the guess
    total = total + (scores[i] - y_hat) ** 2  # miss, squared
error = total / n
```

Check it with Episode 0.1's numbers

Student i	Score y	Guess $\hat{y} = 5x + 40$	Miss $y - \hat{y}$	Miss squared
1	49	45	4	16
2	56	50	6	36
3	65	55	10	100
4	71	60	11	121
5	82	65	17	289

$$\text{Error} = \frac{1}{5}(16 + 36 + 100 + 121 + 289) = \frac{562}{5} = 112.4$$

Your turn: read it, then add it

1

$$\sum_{i=1}^3 i$$

2

$$\sum_{i=1}^4 2i$$

3

$$\sum_{i=1}^3 (y_i - 40)$$

Pause the video. Say each one out loud, then work it out.

Letters side by side mean multiply: $2i = 2 \times i$. For #3, use the scores: 49, 56, 65.

Answers

$$\sum_{i=1}^3 i = 1 + 2 + 3 = 6$$

“the sum from i equals 1 to 3 of i”

$$\sum_{i=1}^4 2i = 2 + 4 + 6 + 8 = 20$$

“the sum from i equals 1 to 4 of two i”

$$\sum_{i=1}^3 (y_i - 40) = 9 + 16 + 25 = 50$$

“the sum from i equals 1 to 3 of the quantity y sub i minus 40”

You can now read real AI formulas

$$\theta_{\text{new}} = \theta - \eta \cdot \text{steepness}$$

“theta new equals theta minus eta times steepness”

Ep 0.1’s learning rule; each knob steps by its own steepness

$$z = \frac{x - \mu}{\sigma}$$

“z equals x minus mu, over sigma”

how many spreads (σ) away from average you are

$$\hat{y} = \sum_{j=1}^d w_j x_j + b$$

“y-hat equals the sum from j equals 1 to d of w sub j times x sub j, plus b”

Ep 0.1’s line with d inputs: the first step of an artificial neuron

THE EQUATION OF THE DAY

$$\sum_{i=1}^n x_i = x_1 + x_2 + x_3 + \cdots + x_n$$

Sigma means: start at the bottom, count up to the top, and add up everything along the way.

NEXT TIME • EPISODE 0.3

Functions as machines

Input goes in, output comes out.

YOUR CHALLENGE

Answer in Episode 0.3

$$\sum_{i=1}^5 x_i y_i$$

Say it out loud. Work it out by hand. Then check it with the Python file (steps at the bottom).